

AI Model Card

NovaTel UK: Customer Churn Prediction & Retention Engine

DOCUMENT TYPE AI Model Card	PROJECT AI Governance & Risk Management	VERSION 1.0
MODEL OWNER Kevin Wadsworth	SYSTEM Customer Churn Prediction Engine	CLASSIFICATION Portfolio / Demonstration

EXECUTIVE SUMMARY	APPROVED WITH CONTROLS
This model is classified high-risk under the EU AI Act (Annex III). Initial fairness testing identified a disparate impact violation (ratio 0.516) in churn-flag rates across age bands; per-group threshold calibration resolved this to 0.987, clearing the 80% rule. Human oversight is enforced via a £50/month auto-approval limit. Approved for use with these controls in place, subject to quarterly fairness monitoring.	

1. Model Overview

Predicts customer churn likelihood for NovaTel UK using behavioural and account features, triggering automated retention offers or routing to human review above a defined value threshold. Classified as high-risk under the EU AI Act (Annex III: profiling with material effect on contract terms).

2. Intended Use

Intended solely to prioritise retention outreach and inform offer eligibility for existing NovaTel customers. Not intended for credit decisions, employment screening, or any use outside customer retention. Any offer or contract change above £50/month requires human sign-off before action is taken.

3. Training Data

6,000 synthetic customer records generated to reflect realistic telecom usage patterns: tenure, contract type, monthly charge, usage volume, support ticket history, payment method and age band. Data is synthetic for portfolio purposes; a production deployment would require an equivalent data protection impact assessment against real customer data before training.

4. Model Type & Performance

Logistic regression classifier (scikit-learn), trained with class-balanced weighting on a 75/25 train/test split. Recall is prioritised over precision by design, since missing an at-risk customer (false negative) is judged more costly than an unnecessary retention offer (false positive). Test-set results below.

ACCURACY	PRECISION	RECALL	F1 SCORE	ROC AUC
0.627	0.215	0.615	0.319	0.685

Test set: n = 1,500 held-out customers (25% split), evaluated after training on the remaining 4,500.

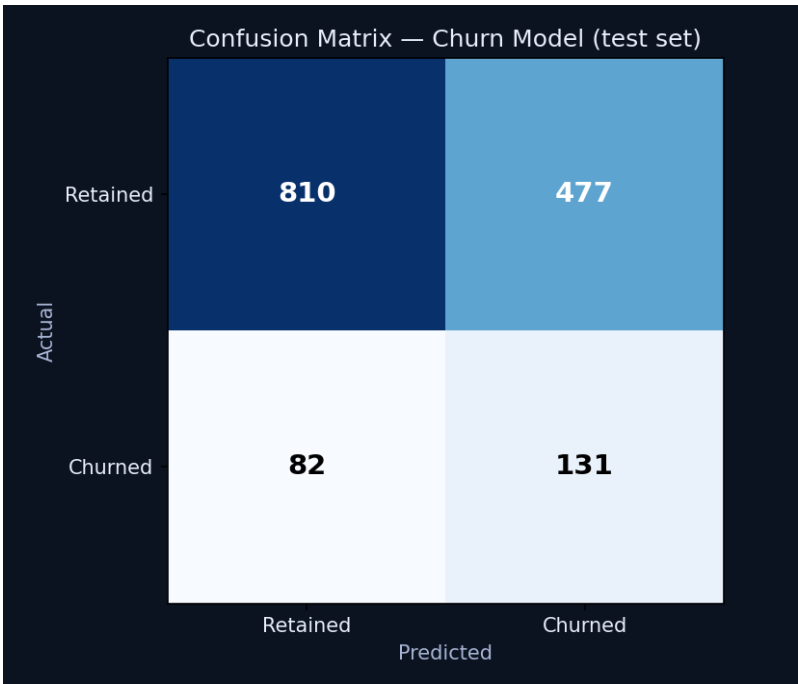


Figure 1: Confusion matrix on the held-out test set (n=1,500).

4. Model Type & Performance (continued)

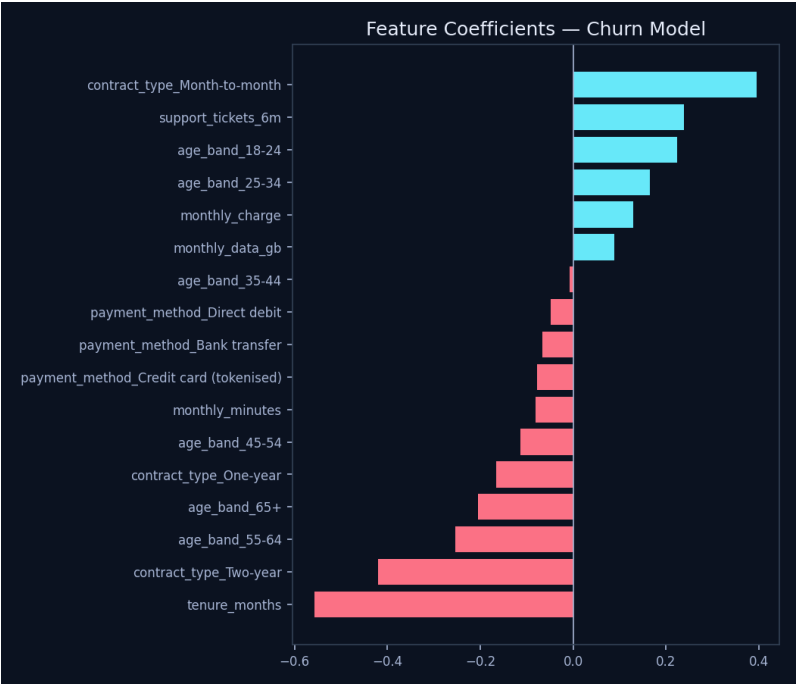


Figure 2: Model coefficients. Age bands 18-24 and 25-34 carry a positive coefficient, meaning the model treats younger age directly as a churn-risk signal, flagged as a fairness finding below.

5. Fairness Testing

A post-hoc fairness check compared churn-flag rates across age bands (a protected characteristic under the UK Equality Act 2010, used here as a proxy grouping variable). Before mitigation, the 18-24 band was flagged at 53.6% versus 27.6% for the 55-64 band, a disparate impact ratio of 0.516, failing the standard four-fifths (80%) rule commonly used to screen for adverse impact.

FINDING: FAILS 80% RULE

Disparate impact ratio 0.516 (threshold: 0.80). The 18-24 age band was flagged at nearly double the rate of the 55-64 band for an otherwise equivalent risk profile.

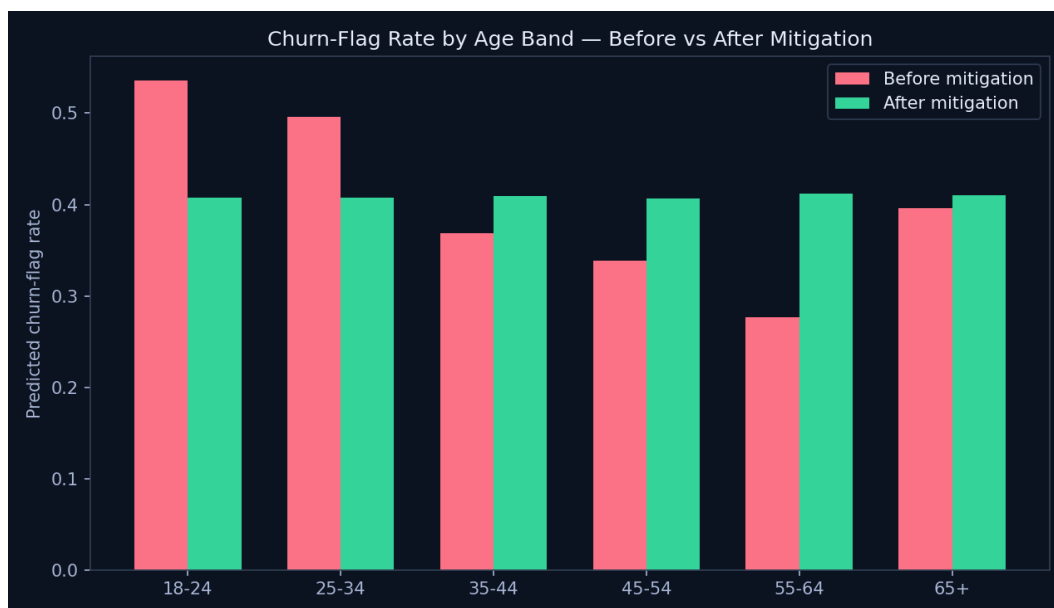


Figure 3: Churn-flag rate by age band, before and after mitigation.

6. Mitigation Applied

Per-group threshold calibration: rather than one global decision threshold, each age band's threshold was calibrated so that its flag rate converges toward the overall base rate (40.5%). This is a recognised post-processing fairness technique. After mitigation, the disparate impact ratio improved from 0.516 to 0.987, comfortably clearing the 80% threshold. This maps directly to risk AI-001 in the project's risk register (inherent score 6, residual score 3).

RESULT: PASSES 80% RULE

Disparate impact ratio improved to 0.987 (threshold: 0.80) after per-group threshold calibration. Residual risk score reduced from 6 to 3 (risk AI-001).

7. Known Limitations

(1) Training data is synthetic, not real customer data; real-world deployment requires retesting against actual distributions. (2) Logistic regression was chosen for interpretability over raw predictive power; a more complex model may need separate fairness testing. (3) Age band is used as a proxy for fairness testing purposes; a full deployment would also test tenure, contract type interactions and any other characteristic with a plausible proxy relationship to protected groups. (4) Per-group threshold calibration mitigates disparate flag rates but does not by itself change the underlying feature weighting. A longer-term fix would involve retraining without age as a direct input, or with a fairness-constrained objective.

8. Human Oversight

All retention offers or contract changes above £50/month, and any full cancellation-risk flag, are hard-blocked from automatic execution and routed to a human reviewer (EU AI Act Art. 14; GDPR Art. 22 right to human review of automated decisions).

9. Monitoring Plan

Disparate impact ratio recalculated quarterly against live flag data. Model accuracy monitored monthly; retraining triggered if accuracy drops more than 5 percentage points month-on-month. Full details tracked in the live AI risk register and dashboard.

Appendix A: Risk Register Cross-Reference

How this document's sections relate to the live AI risk register.

Risk ID	Risk	Addressed In
AI-001	Model under-scores churn risk for a protected segment	Sections 5, 6 (Fairness Testing & Mitigation)
AI-002	High-value offer issued without human review	Section 8 (Human Oversight)
AI-003	Training data drift degrades score accuracy	Section 9 (Monitoring Plan)
AI-004	Customer cannot obtain explanation for a negative outcome	Not directly covered; see AI Governance case study, Section 7
AI-005	Payment metadata inadvertently expands PCI scope	Out of scope for this document; see AI Governance case study, Section 4
AI-006	Model reused for a new purpose without re-assessment	Section 2 (Intended Use)
AI-007	Third-party enrichment source introduces unvetted PII	Not applicable to this synthetic dataset; tracked for future real-data integration
AI-008	Model occasionally misjudges edge-case customers	Section 7 (Known Limitations)

Appendix B: Version History

Version	Date	Author	Change Description
1.0	2026-09-15	Kevin Wadsworth	Initial model card: training results, fairness testing finding, mitigation applied.

Document Approval

Role	Name	Date	Signature
Prepared by	Kevin Wadsworth	2026-09-15	—
Reviewed by	AI Governance Committee	Pending	—
Approved by	Pending	Pending	—

Kevin Wadsworth: AI Governance & Risk Management portfolio project. Model, dataset and evaluation code available in the project repository.